

Practice Quiz: Agents

Part A: Multiple Choice

1. In Active Inference, an agent is distinguished from a passive system by: A) Having a nervous system B) Having active states that influence the environment to minimize free energy C) Being made of organic material D) Being able to communicate with other agents
2. Spinoza's concept of *conatus* refers to: A) The drive to maximize pleasure and minimize pain B) The striving of every being to persist in its own existence C) The tendency of systems to increase in complexity D) The moral obligation to help others persist
3. According to Metzinger, a self-model is: A) A photograph of oneself B) An internal representation of oneself as a distinct entity C) A philosophical impossibility D) A model that only humans can possess
4. The "problem of other minds" in the Active Inference framework becomes: A) Irrelevant, because all systems are agents B) A problem of inferring another system's internal states through one's own generative model C) Unsolvable, because Markov Blankets prevent access to internal states D) A linguistic confusion rather than a genuine philosophical problem
5. Varela's concept of "sense-making" refers to: A) Linguistic meaning production B) An organism's active engagement in generating significance from its environment C) The process of translating raw sense data into propositions D) A computational operation performed only by neural networks
6. If agency exists on a continuum, which philosophical consequence follows most directly? A) Moral status must also be graded rather than binary B) Only the most complex agents have genuine free will C) Simple agents cannot be studied scientifically D) The concept of agency is philosophically useless

7. Dewey's concept of the "organism-environment transaction" anticipated Active Inference by emphasizing: A) That organisms and environments are separate, independent entities B) That cognition is fundamentally about internal computation C) That organism and environment are co-constituted through ongoing interaction D) That evolution is the only form of adaptation

Part B: Short Answer

1. Explain why, in Active Inference, agency does not require consciousness. What philosophical problems does this create?
2. A reinforcement learning algorithm selects actions that maximize expected reward. In what philosophically relevant ways does this differ from an Active Inference agent that minimizes expected free energy?
3. Construct a brief argument for or against the claim: "An ant colony is a single agent with beliefs and goals."